



Title: Climate-Resilient Agrivoltaic Systems: Enhancing Agricultural Sustainability in Southeast Spain

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Abstract:

To respond to the critical challenges facing modern agriculture in the Southeast of Spain—such as climate change, recurring droughts, torrential rain, and excessive solar radiation—we have developed a resilient and sustainable agrivoltaic system. This innovative solution is designed to optimize solar radiation for crop development in real-time while providing a physical shield against extreme weather events. A core objective of this technology is to significantly reduce and eliminate the carbon footprint associated with intensive agricultural activities. By integrating renewable energy production directly into the farming infrastructure, the system powers its own automated irrigation and lighting technologies, replacing fossil fuel dependency with clean, on-site energy. The system features a mobile roof composed of solar panels that move in a coplanar manner to manage light transmission. During periods of peak radiation, the panels close to protect crops from thermal stress; during torrential rains, they close to act as a catchment surface. This allows for the intelligent collection and storage of rainwater for efficient irrigation, preventing the leaching of pollutants into aquifers and reducing surface runoff that contaminates nearby ecosystems. Furthermore, the system is equipped with luminescent spotlights powered by the stored photovoltaic energy to ensure optimal light spectra for plant growth even when the panels are closed. This integrated approach not only enhances climate resilience but also provides a transformative model for carbon-neutral, environmentally respectful horticulture.

Biography

José Miguel Molina-Martínez is a Professor and the current Director of the School of Agricultural Engineering at the Technical University of Cartagena (UPCT). Holding a PhD in Agricultural Engineering, he serves as the Principal Investigator of the "Agromotics and Sea Engineering" (IAM) R&D group. His research focuses on automation, remote control, and the efficient management of water and energy resources, with more than 40 articles published in high-impact journals (JCR Q1) and several patents currently in use. He has led numerous national and international projects centered on technological sustainability within the agri-food sector.

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